

Density matrix

- Write the density matrix for each of the following ensembles. Are the ensembles pure?
 - A completely polarized beam with $S_z +$ (i.e. $|\psi\rangle = |z+\rangle$).
 - A completely polarized beam with $S_x \pm$
 - An unpolarized beam. This can be regarded as an incoherent mixture of a spin-up ensemble and a spin-down ensemble with equal weights (50% each).
 - A partially polarized beam made up of a 75-25 mixture of two pure ensembles, one with $S_z +$ and the other with $S_x +$.

Solution:

Following the notation Prof. Lu has been using in class, the eigenvectors of $\hat{Z}$ are $|0\rangle$ with eigenvalue 1 and $|1\rangle$ with eigenvalue -1 . The eigenvectors of $\hat{X}$ are $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

$$a. \rho = |0\rangle\langle 0| \doteq \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$b. \rho = |\pm\rangle\langle \pm| = \left(\frac{1}{\sqrt{2}}\right)(|0\rangle \pm |1\rangle) \left(\frac{1}{\sqrt{2}}\right)(\langle 0| \pm \langle 1|) \doteq \begin{pmatrix} \frac{1}{2} & \pm \frac{1}{2} \\ \pm \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$c. \rho = \left(\frac{1}{2}\right)|0\rangle\langle 0| + \left(\frac{1}{2}\right)|1\rangle\langle 1| \doteq \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$d. \rho = \left(\frac{3}{4}\right)\rho_{Z+} + \left(\frac{1}{4}\right)\rho_{X+} \doteq \left(\frac{3}{4}\right)\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \left(\frac{1}{4}\right)\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{7}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} \end{pmatrix}$$

To check whether an ensemble is pure, check whether $\rho^2 = \rho$. In the examples above, the first two (a and b) are pure, while the other two (c and d) are not.

Operators

- What is a Hermitian operator? What is a Unitary operator?

Solution: Hermitian means $A = A^\dagger$. Unitary means $AA^\dagger = 1$.

- Which of the following operators are Hermitian? Which are Unitary?

$$(a) \begin{pmatrix} -i & i \\ i & i \end{pmatrix}$$

$$(b) \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

$$(c) \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & i \\ 1 & -i & 3 \end{pmatrix}$$

$$(d) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$(e) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & i & 0 \\ -i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$(f) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution: Hermitian (d), (e), and (f), Unitary (b), (f)

3. Is the product of two Hermitian matrices always Hermitian? If not, under what conditions might the product be Hermitian? Pick two Hermitian operators from the previous problem and check explicitly whether $M_1 M_2$ is Hermitian.

Solution: Not always. Note that this would require that they commute.

$$\hat{A} = \hat{H}_1 \hat{H}_2 = \hat{A}^\dagger = (\hat{H}_1 \hat{H}_2)^\dagger = \hat{H}_2^\dagger \hat{H}_1^\dagger = \hat{H}_2 \hat{H}_1 \Rightarrow \hat{H}_1 \hat{H}_2 - \hat{H}_2 \hat{H}_1 = 0$$

4. Is $e^{-i\hat{A}}$ Hermitian if $\hat{A}$ is Hermitian? If $e^{-i\hat{A}}$ is Unitary, what is the condition on $\hat{A}$?

Solution:

$$\text{No. } (e^{-i\hat{A}})^\dagger = e^{i\hat{A}^\dagger} = e^{i\hat{A}} \neq e^{-i\hat{A}}$$

If $(e^{-i\hat{A}}) (e^{-i\hat{A}})^\dagger = 1$ then $(e^{-i\hat{A}}) (e^{i\hat{A}^\dagger}) = e^{-i(\hat{A} - \hat{A}^\dagger)} = 1$ which is only true if $\hat{A} - \hat{A}^\dagger = 0$, or $\hat{A} = \hat{A}^\dagger$. For $e^{-i\hat{A}}$ to be Unitary, $\hat{A}$ must be Hermitian.

5. When can you replace $e^{-i\hat{H}t/\hbar}$ with $e^{-iE_nt/\hbar}$? Why?

Solution: Only when acting on an energy eigenstate. This would allow us to write

$$e^{-i\hat{H}t/\hbar} |\phi_n\rangle = \left(1 - \frac{i\hat{H}t}{\hbar} - \frac{1}{2} \left(\frac{\hat{H}t}{\hbar} \right)^2 + \dots \right) |\phi_n\rangle = \left(1 - \frac{iE_nt}{\hbar} - \frac{1}{2} \left(\frac{E_nt}{\hbar} \right)^2 + \dots \right) |\phi_n\rangle = e^{-iE_nt/\hbar} |\phi_n\rangle$$

6. Consider the operator:

$$(|+\rangle\langle+|) + (|-\rangle\langle-|)$$

- Explain in what sense this is an operator.
- Explain why this operator is the identity operator.

Solution: a. This is an operator in the sense that it acts on a ket and returns a ket. b. The identity operator acts on an arbitrary ket and returns the same ket. We can test whether this operator is the identity operator by having an arbitrary ket $|\psi\rangle = c_+|+\rangle + c_-|-\rangle$, where c_+ and c_- are constants, and performing the operations:

$$(|+\rangle\langle+|) + (|-\rangle\langle-|) \cdot |\psi\rangle$$

$$(|+\rangle\langle+|) + (|-\rangle\langle-|) \cdot (c_+|+\rangle + c_-|-\rangle)$$

$$(|+\rangle\langle+|) \cdot (c_+|+\rangle + c_-|-\rangle) + (|-\rangle\langle-|) \cdot (c_+|+\rangle + c_-|-\rangle)$$

$$(|+\rangle\langle+|) \cdot c_+|+\rangle + (|+\rangle\langle+|)c_-|-\rangle + (|-\rangle\langle-|) \cdot c_+|+\rangle + (|-\rangle\langle-|)c_-|-\rangle$$

$$c_+|+\rangle\langle+|+\rangle + c_-|+\rangle\langle+|-\rangle + c_+|-\rangle\langle-|+\rangle + c_-|-\rangle\langle-|-\rangle$$

$$c_+|+\rangle(1) + c_-|+\rangle(0) + c_+|-\rangle(0) + c_-|-\rangle(1)$$

$$c_+|+\rangle + c_-|-\rangle = |\psi\rangle$$

7. Why do we use Hermitian operators in quantum mechanics? What property do they have and why do we need it?

Solution: We needed Hermitian operators because they have real eigenvalues. If we want them to correspond to physical observables, then we need a measurement associated with that operator to yield a real number.

8. In the context of quantum mechanics, why are Unitary operators useful? Specifically (a) what can they change? (b) What do they preserve? (c) Is there an analog of this in Cartesian space?

Solution: They are used to perform rotations in Hilbert spaces. (a) They change the projection onto various states, which corresponds to a change of basis. (b) They preserve the magnitude of any state and the expectation value of any operator. This allows you to change bases without changing the expectation value or violating the normalization of states. (c) This is equivalent to rotating your axes in Cartesian space. A vector that initially was along the x direction, may have projections along x' and y' in the rotated axes.

Measurement

1. You are doing homework with three friends: A, B, and C. Each makes a comment about Hermitian matrix operation and measurement. Decide with whom, if anyone, you agree. Discuss this with your neighbor(s).

A says, "Acting with a Hermitian operator $\hat{H}$ is the same thing making a measurement in the lab of the physical observable associated with $\hat{H}$."

B says, "Well, practically. Acting with $\hat{H}$ is just math, whereas measurement is more complex. But acting with $\hat{H}$ and making the measurement always yield the same results (numbers, states, etc.)."

C says, "Acting with operator $\hat{H}$, and making a measurement in the lab are very different things that can yield different numbers, states, etc."

Solution: Friend C is correct. The principal difference between the two processes is that a measurement collapses a system into an eigenstate of the thing being measured. That is, if energy is measured, the final state of the system is an energy eigenstate. That is true whether the system starts out in an energy eigenstate or not. Operator action, on the other hand, can be distributed through to a superposition of eigenstates. Consider an arbitrary Hermitian operator $\hat{O}$ (not necessarily the Hamiltonian) associated with a physical observable O, and having possible eigenvalues $o_1, o_2, \dots$ etc. Consider an initial state which is an eigenstate of this operator: $|\psi_i\rangle = |o_1\rangle$. Then making a measurement of the observable O will yield o_1 , and acting with the operator $\hat{O}$ will yield $\hat{O}|\psi_i\rangle = o_1|o_1\rangle \dots$ very similar indeed. But now consider an initial state which is a mixture of eigenstates of $\hat{O}$: $|\psi_i\rangle = c_1|o_1\rangle + c_2|o_2\rangle$. A measurement will yield either o_1 (with probability $|c_1|^2$) or o_2 (with probability $|c_2|^2$). In contrast, acting with the

operator $\hat{O}$ yields $\hat{O}|\psi_i\rangle = o_1 c_1 |o_1\rangle + o_2 c_2 |o_2\rangle$. This has NOT collapsed the state into one or the other eigenstate but has left the state as a superposition of both.

2. Given your response to the previous question, what is the relationship between operator action and measurement in the lab?

Solution: They are not the same. They are very similar when the system is in an eigenstate of the operator (just the difference between math and lab experiments, but they yield the same results). They are very different when the system is in a mixture of eigenstates of the operator. In that case, the two things can only be connected through the ensemble average. Many measurements will average to the expectation value, which can also be calculated using the operator: $\langle \hat{O} \rangle = \langle \psi_i | \hat{O} | \psi_i \rangle$.

3. Exercise 2.57 from N&C: Cascaded measurements are single measurements
Suppose $\{L_l\}$ and $\{M_m\}$ are two sets of measurement operators. Show that a measurement defined by the measurement operators $\{L_l\}$ followed by a measurement defined by the measurement operators $\{M_m\}$ is physically equivalent to a single measurement defined by measurement operators $\{N_{lm}\}$ with the representation $N_{lm} \equiv \{M_m L_l\}$.

Solution: Suppose the sequence of measurements yields the result l , then the result m . After measuring l , the state of the system is given by

$$|\psi'\rangle = \frac{L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}}$$

where $|\psi\rangle$ is the initial state. Next we measure the result m for state $|\psi'\rangle$. The resulting state is given by

$$\begin{aligned} |\psi''\rangle &= \frac{M_m |\psi'\rangle}{\sqrt{\langle \psi' | M_m^\dagger M_m | \psi' \rangle}} \\ &= \frac{M_m L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}} \frac{\sqrt{\langle \psi | L_l^\dagger L_l | \psi \rangle}}{\sqrt{\langle \psi | L_l^\dagger M_m^\dagger M_m L_l | \psi \rangle}} \\ &= \frac{M_m L_l |\psi\rangle}{\sqrt{\langle \psi | L_l^\dagger M_m^\dagger M_m L_l | \psi \rangle}} \\ &= \frac{N_{lm} |\psi\rangle}{\sqrt{\langle \psi | N_{lm}^\dagger N_{lm} | \psi \rangle}} \end{aligned}$$

Where $N_{lm} \equiv M_m L_l$. Hence, we see that the final state is the same as if the single measurement represented by N_{lm} was applied.

4. Prove that the unitary operator, $e^{-\frac{i\pi}{2n}\hat{Z}}$, can be written as $\cos\left(\frac{\pi}{2n}\right)\hat{\mathbb{I}} - i\sin\left(\frac{\pi}{2n}\right)\hat{Z}$. That is, $e^{-\frac{i\pi}{2n}\hat{Z}} = \cos\left(\frac{\pi}{2n}\right)\hat{\mathbb{I}} - i\sin\left(\frac{\pi}{2n}\right)\hat{Z}$. Hint: act the operator on $|+\rangle$, one of the eigenstates of $\hat{X}$.

Solution: Following the hint and acting this operator on $|+\rangle$

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = e^{-\frac{i\pi}{2n}\hat{Z}}\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \text{ where we have written the state } |+\rangle \text{ in terms of the eigenstates of } \hat{Z}: |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \frac{1}{\sqrt{2}}e^{-\frac{i\pi}{2n}\hat{Z}}|0\rangle + \frac{1}{\sqrt{2}}e^{-\frac{i\pi}{2n}\hat{Z}}|1\rangle$$

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \frac{1}{\sqrt{2}}e^{-\frac{i\pi}{2n}}|0\rangle + \frac{1}{\sqrt{2}}e^{+\frac{i\pi}{2n}}|1\rangle \text{ where we have used the fact that we can substitute the operator by the eigenvalue in a function if the function is operating on an eigenstate of the operator. The eigenvalue of } |0\rangle \text{ is } 1 \text{ and the eigenvalue of } |1\rangle \text{ is } -1.$$

Replace $e^{\pm i\theta} = \cos\theta \pm i\sin\theta$:

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\pi}{2n}\right) - i\sin\left(\frac{\pi}{2n}\right)\right)|0\rangle + \frac{1}{\sqrt{2}}\left(\cos\left(\frac{\pi}{2n}\right) + i\sin\left(\frac{\pi}{2n}\right)\right)|1\rangle$$

Distribute and combine like terms:

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \frac{1}{\sqrt{2}}\cos\left(\frac{\pi}{2n}\right)|0\rangle - \frac{i}{\sqrt{2}}\sin\left(\frac{\pi}{2n}\right)|0\rangle + \frac{1}{\sqrt{2}}\cos\left(\frac{\pi}{2n}\right)|1\rangle + \frac{i}{\sqrt{2}}\sin\left(\frac{\pi}{2n}\right)|1\rangle$$

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \cos\left(\frac{\pi}{2n}\right)\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) - i\sin\left(\frac{\pi}{2n}\right)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

Recognize that $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \cos\left(\frac{\pi}{2n}\right)|+\rangle - i\sin\left(\frac{\pi}{2n}\right)|-\rangle$$

So, the unitary operator acts on the state $|+\rangle$ and turns it into a state with two terms: one that is the state $|+\rangle$, which can be obtained by acting on the state by the identity operator, and one that is the state $|-\rangle$, which can be obtained by acting the operator $\hat{Z}$ since $\hat{Z}|+\rangle = |-\rangle$ (convince yourself).

$$e^{-\frac{i\pi}{2n}\hat{Z}}|+\rangle = \cos\left(\frac{\pi}{2n}\right)\hat{\mathbb{I}}|+\rangle - i\sin\left(\frac{\pi}{2n}\right)\hat{Z}|+\rangle \text{ and } e^{-\frac{i\pi}{2n}\hat{Z}} = \cos\left(\frac{\pi}{2n}\right)\hat{\mathbb{I}} - i\sin\left(\frac{\pi}{2n}\right)\hat{Z}$$